

One-bit encoded reconfigurable intelligent surface-aided radar anti-suppression interference

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ABSTRACT

Reconfigurable intelligent surfaces (RISs) are characterized by low cost and strong survivability and can change the electromagnetic characteristics of signals. Based on these characteristics, this paper deploys an RIS near a monostatic radar. The radar can receive direct signals from the target (jammer) and reflected signals from the RIS, thereby achieving multiview monostatic radar anti-main lobe suppression interference. This paper first establishes an RIS-aided radar echo signal model and then systematically analyzes the ability of the RIS-aided radar to resist main lobe suppression interference. Subsequently, the phase shift vector of the 1-bit encoded RIS is obtained using a genetic algorithm to achieve beam control. The simulation results showed that introducing an RIS to assist monostatic radar can effectively improve the anti-main lobe suppression interference ability of monostatic radar at a relatively low cost.

1. Introduction

Reconfigurable intelligent surfaces (RISs) are low-cost artificial surfaces composed of a periodic arrangement of subwavelength refractive/reflective units that can change the electromagnetic characteristics of incident signals, such as phase, amplitude, frequency, and polarization. Appropriate electronic control devices are used to reshape the signal propagation environment [1–4]. Generally, RIS technologies mainly involve two types of architectures, namely passive RIS and active RIS. Widely used in 5G or 6G communications and sensor systems, passive RISs can alter the propagation environment or reach blind spots [5]. In addition, passive RISs have been successfully applied to dual-function radar communication systems to improve the potential of radar sensing and communication functions [6,7]. However, the passive RIS also has some drawbacks. Due to the signal reflected by the RIS needing to pass through two paths, the received signal suffers from double path loss attenuation and therefore becomes weak enough. In order to address this issue, active RIS has been proposed and applied in communication systems [8–10]. For engineering simplicity, one-bit encoded passive RIS is employed in this paper. We will refer to the 'passive RIS' as RIS in the following sections.

In the field of radar research, an RIS is used for the first time to

reduce the radar scattering cross section (RCS) [11,12]. Recently, RISs are often used to assist in radar target detection [13–16] and target positioning [17–20] owing to their low cost and easy deployment. In [13], a method of using RIS-aided radar for target detection is first proposed. According to the basic model of RIS-aided radar target detection and different target sizes, antenna wavelengths, and other data, the RIS target detection performance improved under two situations, and the relative positions of the radar and RIS, whether near or far, are discussed. The simulation experiments verify that the RIS can change the receiving signal-to-noise ratio (SNR) and that the RIS deployment position can improve the target detection performance. In [14], a far-field RIS is used to expand the field of view of scanning radar and realize radar surveillance in nonlinear-of-sight (NLOS) scenarios. The study in [15] proposes a new system in which forward and backward RIS assist monostatic/multi-static multi-input multi-output (MIMO) radar performs during target detection and the phase shift design problem introduced by the RIS is solved by maximizing the target detection probability under a fixed false alarm probability. In [16], multiple RISs are deployed in a MIMO system to improve clutter suppression and multitarget detection performance by jointly optimizing the space-time transmit waveform, the receiving filter of the radar and the reflection coefficient of the RIS. An RIS-aided millimeter-wave MIMO radar system

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in [17] is proposed for multitarget positioning. A hierarchical codebook is designed for the blocked line of sight between the transmitter and the target, and an adaptive positioning algorithm was used to detect multiple targets with sufficient accuracy. The work in [18] equips the unmanned aerial vehicle (UAV) group with RIS. The target signal is reflected to the central UAV via the RIS equipped with the UAV group. The impact of UAV movement on direction of arrival (DOA) estimation accuracy is removed using atomic norms to achieve high-precision DOA estimation of the target. In [19], a DOA estimation method for a single antenna is proposed. An RIS heterogeneous array and compressed sensing theory are combined to achieve accurate DOA estimation. In [20], an RIS is used to provide a reflected path. Two RIS phase design strategies are proposed to improve the estimation performance of the RIS-aided scheme. The coherence of the signals from the direct path and the RIS reflection path is maximized by optimizing the RIS phase shift. In this proposed scheme, the phase shift for minimizing the CRB of the DOA estimate is obtained via the manifold optimization method.

However, studies on the application of the RIS in the field of anti-interference are rare. Given the ability of RISs to change the electromagnetic characteristics of signals and expand the radar field of view, they may also assist monostatic radars in achieving anti-main lobe suppression interference, indicating their broad application prospects.

Monostatic radar can effectively suppress interference from side lobes via sidelobe blanking and cancellation [21]. During main lobe suppression interference, the target echo received by the radar is masked by the high-power noise interference signal, which in turn causes the radar's performance to be greatly reduced during target detection and tracking. Conventional signal processing methods can no longer suppress main-beam interference. Improving the ability of monostatic radars to resist main-beam suppression interference requires an urgent solution.

In this paper, an RIS-aided monostatic radar anti-suppressive interference method is proposed to suppress main lobe interference. The RIS is deployed near the monostatic radar. The monostatic radar receives signals with dual beams. A beam receives direct channel to the target and interference, while another beam receives the target signal and interference signal reflected by the RIS. This scheme can achieve multiview detection of monostatic radar, thereby utilizing the spatial difference characteristics of the target echo signal and interference signal to achieve interference suppression. For beam control, the genetic algorithm obtains the phase shift vector of the RIS to indicate the main lobe of the RIS in the direction of the target. For different types of main lobe suppression interference, the simulation experiments showed that the proposed method can effectively suppress main-beam suppression interference. Utilizing an RIS to control electromagnetic waves can significantly improve the anti-interference performance of monostatic radar at a low cost.

This paper is organized as follows. Section 2 presents the RIS-aided monostatic radar scenario. Section 3 introduces the design of the phase shift vector of the RIS. The noise subspace projection used for

performance of the RIS-aided monostatic radar against main lobe suppression interference. The conclusions are presented in Section 6.

2. Signal model

We consider an RIS-aided monostatic radar scenario, as shown in Fig. 1. An RIS is deployed near the radar station. Assuming that the radar station is working in the receiving and transmitting status, the RIS is composed of M units arranged in a uniform linear array with an element spacing of half a wavelength. Additionally, for the RIS, only 2 phases (i. e., 0 and 180°) can be controlled by the system. In the radar detection area, a suppressive jammer covers the enemy target. The radar transmitting beam is pointed toward the target and the area where the jammer is located, while two receiving beams are pointed toward the target and RIS. The received signal of the radar station includes two parts. The first part is the direct echo signal reflected from the target (jammer) received by the direct receiving channel. The second part is reflected by the target (jammer) to the RIS and then by the RIS to the radar receiving channel echo signal. The receiving beam direction of the RIS is determined by the amplitude and phase parameters of each reflection unit and is uniformly controlled by the radar. The radar-illuminated region mentioned in the paper is located in the air, allowing clutter to be suppressed through the use of Moving Target Detection (MTD).

In the RIS-aided monostatic radar scenario, the echo signal of the direct receiving channel can be expressed as

$$r_1(t) = \alpha_1 \cdot s_{TX}(t - 2R_{rt}/c) \cdot e^{-j2\pi(2R_{rt}/\lambda)} + \beta_1 \cdot J(t - R_{rj}/c) \cdot e^{-j2\pi(R_{rj}/\lambda)} + w_1(t) \quad (1)$$

where $s_{TX}(t)$ is the baseband transmit signal, $\alpha_1 = \sqrt{\frac{P_t G_t G_r \sigma_1}{(4\pi)^3} \frac{\lambda^2}{R_{rt}^4}}$ is the complex amplitude of the target signal, P_t is the radar transmit power, G_t is the main lobe gain of the radar transmitting antenna beam, G_r is the main lobe gain of the radar receiving antenna beam, and σ_1 is the RCS of the target. λ is the wavelength, and R_{rt} is the distance between the radar and the target. $J(t)$ is the baseband interference signal like accompanying jamming or self-defense jamming, $\beta_1 = \sqrt{\frac{P_j G_j G_r \lambda^2}{(4\pi)^2 R_{rj}^2}}$ represents the complex amplitude of the interference signal, P_j is the transmit power of the jammer, G_j is the Antenna gain of the jammer, and R_{rj} is the distance from the radar to the jammer. $w_1(t)$ is the thermal noise of the radar receiver, and is assumed to be Gaussian white noise with a zero mean and variance σ_w^2 .

As the propagation of the detection signal through the Radar-RIS-target-RIS-radar link results in a very high level of signal attenuation, this particular link is not considered. The echo signal of the indirect receiving channel is superposed by the reflected signals of M units of the RIS and can be expressed as [22]

$$r_2(t) = \underbrace{\sum_{m=1}^M e^{j\frac{2\pi\sin\theta_2}{\lambda}(m-1)} d_{e^{j\varphi_m}} e^{j\frac{2\pi\sin\theta_1^T}{\lambda}(m-1)} d_{\alpha_2} s_{TX}(t - (R_{rt} + R_{ri} + R_{ir})/c) \cdot e^{-j2\pi((R_{rt} + R_{ri} + R_{ir})/\lambda)}}_{\text{Radar} \rightarrow \text{Target} \rightarrow \text{RIS} \rightarrow \text{Radar}} + \underbrace{\sum_{m=1}^M e^{j\frac{2\pi\sin\theta_2}{\lambda}(m-1)} d_{e^{j\varphi_m}} e^{j\frac{2\pi\sin\theta_1^T}{\lambda}(m-1)} d_{\beta_2} J(t - (R_{ji} + R_{ir})/c) \cdot e^{-j2\pi((R_{ji} + R_{ir})/\lambda)}}_{\text{Jammer} \rightarrow \text{RIS} \rightarrow \text{Radar}} + w_2(t) \quad (2)$$

main lobe interference suppression is discussed in Section 4. Section 5 presents the different simulations implemented to assess the

where θ_2 is the angle of the RIS relative to the radar, $e^{j\varphi_m}$ is the phase parameter adjusted by the m unit of the RIS, $\varphi_m \in \{0, \pi\}$ and θ_1^T is the

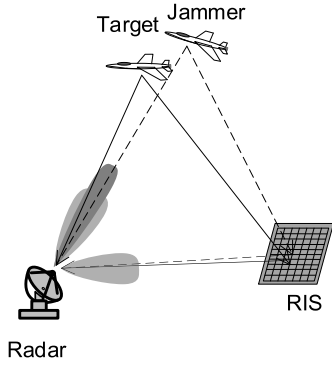


Fig. 1. Schematic of RIS aided radar against suppression jamming.

angle of the target relative to the RIS, $\alpha_2 = \sqrt{\frac{P_t G_t G_r \sigma_2 \zeta_{RIS} \lambda^2}{(4\pi)^4 R_{rt}^2 L_{loss} R_{it}^2 R_{ir}^2}}$ is the complex amplitude of the target signal received by the indirect channel, σ_2 is the bistatic RCS of the target, ζ_{RIS} is the bistatic RCS of the RIS, L_{loss} is the loss factor, R_{it} and R_{ir} are the distances between the RIS, the target, and the radar, respectively. θ_1^j is the angle of the interference relative to the RIS. $\beta_2 = \sqrt{\frac{P_j G_j G_r \zeta_{RIS} \lambda^2}{(4\pi)^4 L_{loss} R_{ji}^2 R_{ir}^2}}$ represents the complex amplitude of the interference signal received by the indirect channel. R_{ji} is the distance between the RIS and the jammer, $w_2(t)$ is the thermal noise of the radar receiver, and is assumed to be Gaussian white noise with a zero mean and variance σ_w^2 .

The purpose of using the RIS-aided radar is to suppress main lobe interference by utilizing the weak correlation of target signals and the strong correlation of interference signals between different channels. The stronger the correlation is between the interference signals received by different receiving channels, the better the interference suppression effect. According to Eqs. (1) and (2), the correlation coefficient between

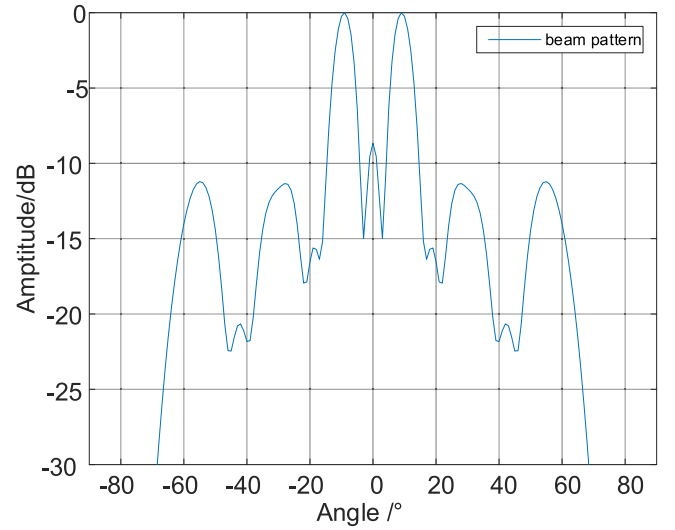


Fig. 2. RIS beamforming with encoding sequence 000011111000000.

the interference signal of the direct receiving channel and the interference signal of the indirect receiving channel can be expressed as

$$\rho_{12}(\tau) = \frac{E[J_1(t)J_2^*(t+\tau)]}{\sqrt{\text{var}[J_1(t)]\text{var}[J_2(t)]}} \quad (3)$$

where $J_1(t) = \beta_1 \cdot J(t - R_{rj}/c) \cdot e^{-j2\pi(R_{rj}/\lambda)}$ is the interference signal of the direct receiving channel. $J_2(t) = \sum_{m=1}^M e^{\frac{2\pi \sin \theta_2}{\lambda}} (m-1)d e^{j\varphi_m} e^{\frac{2\pi \sin \theta_1^j}{\lambda}} (m-1)d \beta_2 \cdot J(t - (R_{ji} + R_{ir})/c) e^{-j2\pi((R_{ji} + R_{ir})/\lambda)}$ represents the interference signal of the indirect receiving channel.

Let $A_m = e^{\frac{2\pi \sin \theta_2}{\lambda}} (m-1)d e^{j\varphi_m} e^{\frac{2\pi \sin \theta_1^j}{\lambda}} (m-1)d$. Thus, Eq. (3) can be written as:

$$\begin{aligned} \rho_{12}(\tau) &= \frac{\beta_1 \beta_2^* e^{j2\pi((R_{ji} + R_{ir} - R_{rj})/\lambda)} E[J(t - R_{rj}/c) \cdot J^*(t + \tau - (R_{ji} + R_{ir})/c)] \sum_{m=1}^M E[A_m e^{-j\varphi_m}]}{|\beta_1| |\beta_2^*| \sqrt{\text{var}[J(t)]} \cdot \sqrt{\text{var}\left[\left(\sum_{m=1}^M A_m e^{-j\varphi_m} J(t - (R_{ji} + R_{ir})/c)\right)\right]}} \\ &= \frac{\beta_1 \beta_2^* e^{j2\pi((R_{ji} + R_{ir} - R_{rj})/\lambda)} E[J(t - R_{rj}/c) \cdot J^*(t + \tau - (R_{ji} + R_{ir})/c)] \cdot \sum_{m=1}^M A_m e^{-j\varphi_m}}{|\beta_1| |\beta_2^*| \left| \sum_{m=1}^M A_m e^{-j\varphi_m} \right| \text{var}[J(t)]} \end{aligned} \quad (4)$$

Table 1

Genetic algorithm for solving RIS phase shift vector.

Input: the angle of target relative to RIS, the vector of the target response, number of iterations, number of populations, crossover probability, mutation probability, array manifold matrix.
Output: w
Repeat
1: Generate S parent individuals randomly.
2: Calculate the fitness value (non-negative, maximized) of each individual:
$f_i = C_{\max} - \ y - Aw\ _2^2$
3: Calculate the sum of fitness values and the selection probability of each individual:
$F = \sum_{i=1}^S f_i$ $C_{\max} - \ y - Aw\ _2^2$
$P(i) = \frac{f_i}{F}$
Use the wheel selection method to repeatedly select individuals with high fitness values to further optimize the population and eliminate individuals with low fitness values.
4: Generate a new population through gene exchange between individuals.
5: Mutate to avoid local optimal solutions.
until convergence

According to Eq. (4), although the propagation attenuation from the jammer to the two receiving channels varies, which results in different complex amplitude powers of the interference signals on a variety of receiving channels, this phenomenon does not affect their correlation. The correlation between interference signals from different receiving channels is affected by the propagation delay difference. When $\tau = (R_{ji} + R_{ir} - R_{rj})/c$, the correlation coefficient is the largest. Therefore, the distance compensation value $\hat{\tau}$ can be estimated by maximizing the cross-correlation coefficient. Delay compensation can be applied to the signals of the two receiving channels to improve the correlation of the interference signals between these two channels.

3. RIS phase modulation

For a 1-bit encoded RIS, the reflective unit is heterogeneously integrated with a passive meta-surface structure and a switching diode. The switching diode has an on and off state. The reflection phase of the unit in the two states has a phase difference of 180° . The numbers “0” and “1” are used to digitize the unit to form a 1-bit phase encoding [23]. By designing the coding sequence of the “0” unit and the “1” unit, the electromagnetic waves irradiated at different positions on the meta-surface are reflected and carry different phases, thereby achieving beamforming of the RIS.

The RIS phase distribution is crucial for enhancing the performance of the proposed system; it can also exploit as much of the RIS potential as possible.

We solve the following optimization problem [24] to obtain the phase shift vector of the RIS so that the main beam of the RIS points in the direction of the target:

$$\begin{aligned} \min & \| \mathbf{y} - \mathbf{A}\mathbf{w} \|^2_2 \\ \text{s.t. } & \varphi_m \in \{0, \pi\} \end{aligned} \quad (5)$$

where $\mathbf{y}$ is the vector of the target response, $\mathbf{A} = [a(\theta_1), a(\theta_2), \dots, a(\theta_N)]$ is array manifold matrix, $a(\theta_i) = [1, e^{j2\pi d \frac{\sin\theta_i}{\lambda}}, \dots, e^{j2\pi d(M-1) \frac{\sin\theta_i}{\lambda}}]$, and φ_m is the unit of the phase shift vector $\mathbf{w} = [e^{j\varphi_1}, \dots, e^{j\varphi_m}]$.

The value of each unit of the solution to the abovementioned optimization problem is limited to 0 or pi and thus can be converted into binary mode. Furthermore, the genetic algorithm [25] is used to search for the optimal encoding mode. As the genetic algorithm requires the fitness function to be nonnegative, the fitness function is set to a constant minus the objective function. The genetic algorithm code is summarized in Table 1.

By assigning specific encoding sequences to the RIS, it becomes possible to control the phase shift of each unit, thereby enabling the RIS to steer its beam towards different directions. When the angle of the target relative to the RIS is 9° , the phase shift vector can be obtained. The beam control of the RIS is shown in Fig. 2. A high-gain beam is formed at the corresponding angle of the target. Fig. 2 depicts the encoding sequence corresponding to a 16-element linear array, where the phase shift of 0° corresponds to the encoding “0”, and the phase shift of 180° corresponds to the encoding “1”.

4. Interference suppression based on orthogonal projection

The aligned signal can be expressed as

$$\mathbf{r}(t) = [\mathbf{r}_1(t), \mathbf{r}_2(t + \hat{\tau})]^T = \mathbf{S}(t) + \mathbf{J}(t) + \mathbf{w}(t) \quad (6)$$

where $\mathbf{S}(t)$, $\mathbf{J}(t)$ and $\mathbf{w}(t)$ are the target echo signal vector, interference signal vector, and noise signal vector respectively. $\mathbf{S}(t)$ and $\mathbf{J}(t)$ can be expressed as:

Table 2
Simulation Settings.

Parameters	Denotations	Values
transmit power	P_t	50 kW
wavelength	λ	0.1 m
sampling frequency	f_s	12 MHz
pulse repetition frequency	f_r	1 KHz
antenna gain	G_t	30 dB
pulse width	τ	20 μ s
bandwidth	B	10 MHz
the number of RIS units	M	16

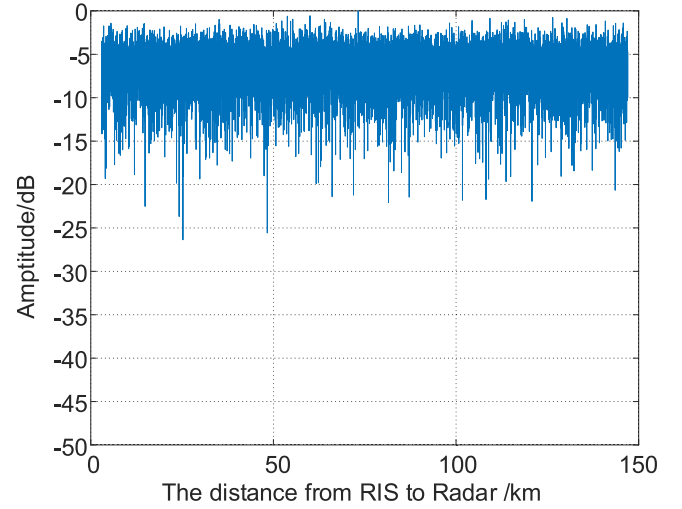


Fig. 3. The output result of pulse compression of the direct channel echo with RF noise jamming.

$$\mathbf{S}(t) = \begin{bmatrix} \alpha_1 \cdot s_{TX} \left(t - \frac{2R_{rt}}{c} \right) \cdot e^{-j2\pi \left(\frac{2R_{rt}}{\lambda} \right)} \\ \sum_{m=1}^M e^{j \frac{2\pi \sin\theta_2}{\lambda} (m-1)d} e^{j\varphi_m} e^{j \frac{2\pi \sin\theta_1}{\lambda} (m-1)d} \\ \cdot \left[\alpha_2 \cdot s_{TX} \left(t + \frac{(R_{ji} - R_{rj})}{c} - \frac{(R_{rt} + R_{rl})}{c} \right) \cdot e^{-j2\pi \left(\frac{R_{rt} + R_{rl} + R_{ir}}{\lambda} \right)} \right] \end{bmatrix} \quad (7)$$

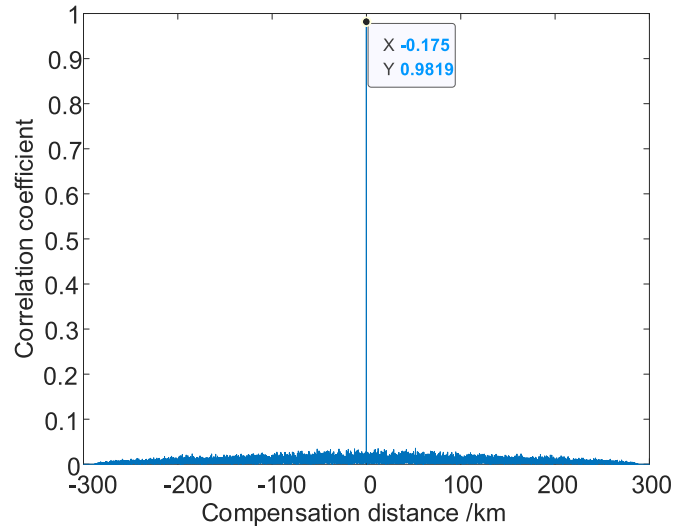


Fig. 4. Propagation delay estimation.

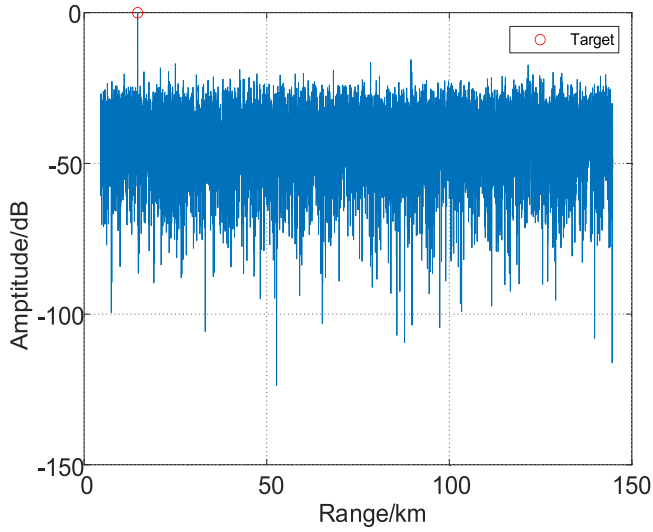


Fig. 5. Output of RF noise jamming suppression.

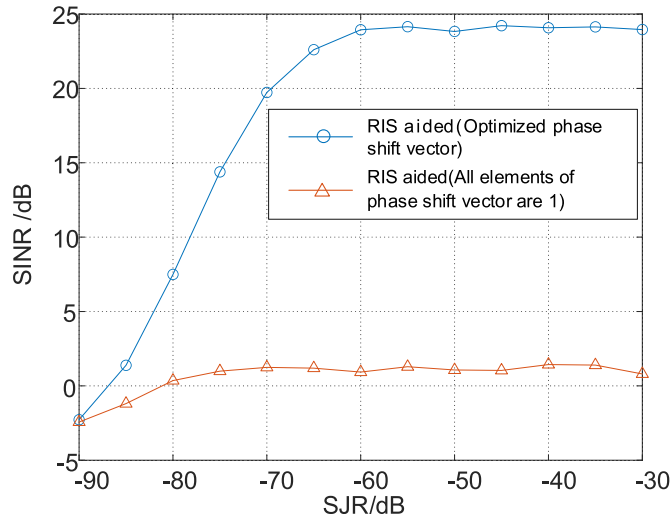


Fig. 6. Output SINR under different SJRs.

$$\mathbf{J}(t) = \begin{bmatrix} \beta_1 \cdot J\left(t - \frac{R_{rj}}{c}\right) \cdot e^{-j2\pi \frac{R_{rj}}{\lambda}} \\ \sum_{m=1}^M A_m \beta_2 \cdot J\left(t - \frac{R_{rj}}{c}\right) e^{-j2\pi \frac{(R_{ji} + R_{jr})}{\lambda}} \end{bmatrix} \quad (8)$$

The covariance matrix of the aligned signal can be expressed as

$$\mathbf{R}(t) = E[\mathbf{r}(t)\mathbf{r}^H(t)] \quad (9)$$

Given that $\mathbf{R}(t)$ is unknown in practice, we can use only the received signal to estimate its parameters and calculate its estimated value, which can be expressed as:

$$\hat{\mathbf{R}} = \sum_{n=1}^N \mathbf{r}(n)\mathbf{r}^H(n) \quad (10)$$

where N is the number of time-domain sampled samples of the received signal. As the power of the suppressed interference signal is much greater than the power of the target echo signal, the feature decomposition of $\hat{\mathbf{R}}$ can be represented as

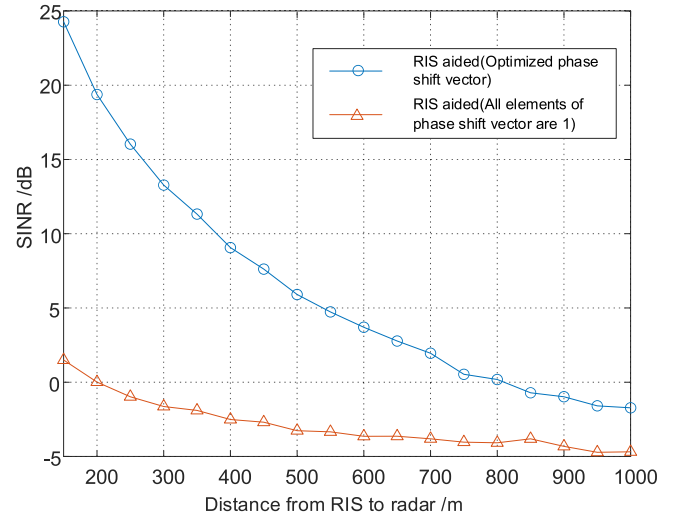


Fig. 7. Output SINR under different distances from RIS to Radar.

$$\hat{\mathbf{R}} = (\sigma_f^2 + \sigma_w^2)\mathbf{u}_1\mathbf{u}_1^H + \sigma_w^2\mathbf{u}_2\mathbf{u}_2^H. \quad (11)$$

Therefore, the subspace formed by the eigenvector corresponding to the first largest eigenvalue $\mathbf{u}_1$ of $\hat{\mathbf{R}}$ is the interference subspace $\mathbf{P}_j = \mathbf{u}_1\mathbf{u}_1^H$. Moreover, the energy of the interference signal is distributed within this subspace. The subspace formed by the eigenvector corresponding to the small eigenvalue $\mathbf{u}_2$ is called the orthogonal complementary space of the interference subspace $\mathbf{P}_j^\perp$. Consequently, the projection matrix projected into subspace $\mathbf{P}_j^\perp$ can be expressed as

$$\mathbf{P}_j^\perp = \mathbf{I} - \mathbf{u}_1\mathbf{u}_1^H. \quad (12)$$

Then, we project the aligned signal into subspace $\mathbf{P}_j^\perp$. Subspace $\mathbf{P}_j$ is orthogonal to subspace $\mathbf{P}_j^\perp$, $\mathbf{P}_j^\perp \mathbf{J}(t) = \mathbf{0}$. Therefore, the signal after interference suppression $\mathbf{z}(t)$ can be expressed as

$$\mathbf{z}(t) = \mathbf{P}_j^\perp \mathbf{r}(t) = \mathbf{P}_j^\perp (\mathbf{S}(t) + \mathbf{w}(t)). \quad (13)$$

5. Simulation results

We consider the system as shown in Fig. 1, which consists of an RIS with coordinates (0, -150)m and a radar with coordinates (0, 0)m. Let the two receiving beams of the radar be pointed at the target (jammer) and RIS. The transmitting antenna of the jammer has the same gain for the radar station and the RIS. The technical parameters used in the simulations are given in Table 2.

5.1. Radio frequency noise interference suppression

The simulation interference type is radio frequency (RF) noise interference. Let the input SNR of the direct channel of the radar station be 15 dB and the signal-to-interference ratio (SJR) be -40 dB. Fig. 3 shows the output signal of the direct channel after pulse compression when the radar station is interfered with. The signal completely blocks the target echo signal, which hinders target detection and other tasks.

Fig. 4 shows the relationship between the correlation coefficient of the two-channel interference signal and the distance compensation. When the distance compensation is 0.175 km, the correlation of the echo signals between the two channels is the highest. Fig. 5 shows the results of the subspace projection. After interference suppression, the signal-to-interference-to-noise ratio (SINR) is approximately 24 dB. We can clearly see the echo signal of the target. Therefore, the method proposed in this paper can effectively suppress RF noise interference signal.

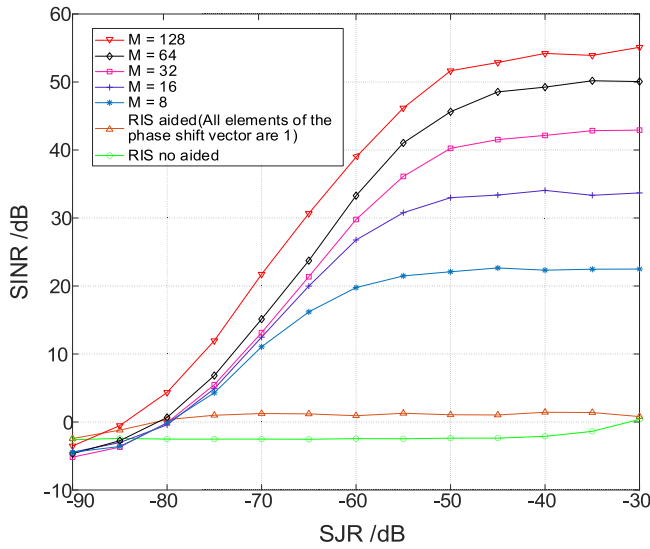


Fig. 8. Output SINR under different numbers of RIS units.

5.2. Simulation of output SINR under different SJRs

A simulation is performed to illustrate the impact of the SJR. Fig. 6 shows the output SINR under different SJRs after interference suppression when the RIS is modulated by the phase shift vector and without the phase modulation. The solution without phase modulation means that the units of the phase shift vector are all 1. Since no corresponding operation is taken on the signal, it only plays the role of a conventional reflection. It can be seen that under the same SJRs, the output SINR after the interference suppression of the echo signal via the RIS phase modulation is higher. Due to the phase modulation of RIS, the main lobe of its beam is directed towards the direction of the target/jammer, resulting in a higher power of the received target/jammer signal, further making the interference signal of the indirect channel have stronger correlation with the interference signal of the direct channel and better interference suppression effect. By contrast, as shown in Fig. 6, the design of the RIS phase shift vector is crucial to the performance of the RIS-aided radar system. Despite the assistance of the RIS, no significant SINR gain can be obtained without optimizing the phase shift vector.

5.3. Simulation of output SINR under different distances from RIS to radar

This simulation is to verify the impact of the distance from the RIS to the radar in terms of improving the output SINR. Fig. 7 shows the SINR after interference suppression when the RIS is modulated by phase shift and without phase shift modulation. The simulated interference type uses RF noise interference. The radar direct channel input SJR is approximately -40 dB. Fig. 7 also shows that at the same distance, the SINR after interference suppression via RIS phase modulation is much greater. As the distance from RIS to radar gradually increases, SINR significantly decreases. The reason for the decrease in the SINR is that the greater the distance from the RIS to the radar, the greater the multiplicative fading of the RIS-reflected signal, and the greater the indirect channel. The weaker the interference signal of the indirect channel is, the worse the interference suppression effect. Considering the signal attenuation caused by multiplicative fading, an RIS should be deployed close to the radar station.

5.4. Simulation of output SINR under different RIS unit numbers

Another simulation is conducted to evaluate the impact of the number of RIS units. Similar to the previous experiment, in addition to

the two cases of whether the RIS phase shift is modulated or not, the case of the radar working alone has been added for comparison. As shown in Fig. 8, as the number of RIS elements increases from 8 to 128, the introduction of the RIS into the scene leads to increasingly significant improvements in the SINR. When the number of RIS elements reaches 128, the system's SINR gain compared to the scenario without RIS assistance reaches 53 dB. Furthermore, it can be observed from the Fig. 8 that the design of RIS reflection coefficients is crucial for the performance of RIS-aided radar. Despite the assistance of RIS, without optimization of phase shift vectors, significant SINR gains cannot be achieved.

6. Conclusion

This paper employs an RIS to assist a monostatic radar. The radar station can receive the direct signal and the signal reflected by the RIS at the same time to achieve multiview detection. A genetic algorithm is used to solve the phase shift vector of the 1-bit encoded RIS to realize beam control. Based on the strong correlation characteristics of interference signals between different channels, interference suppression via orthogonal projection is realized. Our analysis has shown that 1-bit encoded RIS-aided monostatic radar can significantly improve the anti-interference performance of a monostatic radar at low cost. In addition, the RIS should avoid being placed far away from the radar to reduce the multiplicative loss of the received signal in the indirect channel.

Although the proposed method performs well under single suppression interference, aligning interference signal becomes challenging as the number of interfering devices increases, particularly in the presence of compound interference. Therefore, how to utilize high-resolution multi-bit RISs to improve the method in this case will be our subsequent work. Additionally, the introduction of an active RIS for signal amplification will be explored as an extension of this paper, considering that path loss may lead to the attenuation of the target echo energy.

CRediT authorship contribution statement

Yujia Zhang: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing, Formal analysis. **Yu Zhou:** Methodology, Writing – review & editing, Funding acquisition. **Zhanyue Chen:** Validation, Methodology. **Song Shang:** Investigation, Validation. **Linrang Zhang:** Funding acquisition, Supervision. **Lan Du:** Funding acquisition, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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